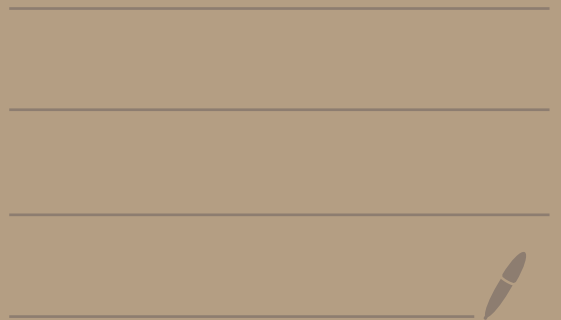


Week 4, Lecture 2

Feb 15, 2024.



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Pumping Lemma. used to prove

languages are not regular.

$$F = \{ ww \mid w \in \{0,1\}^* \}$$

Need to show that F is not regular.

$$F = \{ \epsilon, 00, 11, 0101, 1111, 0000, \dots \}$$

$$0 \notin F, 1 \notin F, 10 \notin F, 110 \in F$$

Proof that F is not regular. (using PL).

- Assume F is regular. Let p be the pumping length.
- Now we need to find some string s in F of length at least p .

$$s = 0^p 1 0^p$$

s into three pieces.

$$s = xyz$$

$$|s| = 2p + 2 \geq p$$

- 1) $\forall i \geq 0, xy^i z \in F$
- 2) $|y| > 0$
- 3) $|xy| \leq p$

$$S = 0^p 1 0^p,$$

$$S = xyz.$$

$$x = 0^{p-1}, \boxed{y = 0}, z = 1 0^p$$

$$1) \forall i \geq 0, xy^i z \in F$$

$$2) |y| > 0 \quad \checkmark$$

$$3) |xy| \leq p \quad \checkmark$$

$$\text{for } i = \overbrace{0, 1, 2, 3, \dots}^{\text{pump up}}$$

$\leftarrow$ PD

$$i=0$$

$$y=0$$

$$y^0 \Rightarrow \epsilon$$

$$xy^0z = 0^{p-1} \epsilon 1 0^p \in F?$$

$$= \underbrace{0^{p-1}}_w \underbrace{1 0^p}_{\neq w} \notin F$$

$$i=2$$

$$0^{p-1} 00 1 0^p$$

$$\equiv$$

$$\underbrace{0^{p+1}}_w \underbrace{1 0^p}_{\neq w}$$

$$i=3, 4, 5, \dots$$

$$\Rightarrow$$

F is not regular.



$$D = \{ 1^{n^2} \mid n \geq 0 \}$$

$$\Sigma = \{1\}$$

$$D = \{ \varepsilon, 1, 1111, 11111111, \dots \}$$

$n=0 \quad n=1 \quad n=2, \quad n=3, \quad \dots$

Task is to show that D is not regular.

Proof using PL.

- Assume D is regular that p is the pumping length.
- Need to find some string $s \in D$ s.t. $|s| \geq p$.

$$s = 1^p \in D ?$$

$$1^{np}$$

$$s = 1^{p^2} \in D \quad \checkmark$$

Can we divide s into three parts?

$$X \quad x = 1^{p-1}, \quad y = 1, \quad z = 1^p$$

$$p-1 + 1 + p = \boxed{2p \neq p^2}$$

Suppose we are able to divide S into tree parts.

$$1) \forall i \geq 0 \quad xy^i z \in D.$$

$$2) |y| > 0$$

$$3) |xy| \leq p$$

$$\text{imagine } |y| \geq 1$$

$$|s| = |xyz| = p^2$$

$$|xy^2z| =$$

will it be a perfect square?

$$|xy^2z| = p^2 + \underline{|y|}$$

$$|y| \leq p$$

$$\leq p^2 + p$$

$$p^2 + p \neq p^2 + 2p + 1 = (p+1)^2$$

not a perfect square

$\Rightarrow D$ is not regular.